Scalability & Future Plan

Date	6 July 2026
Team ID	
Project Name	A comprehensive measure of well being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the HDI Predictor solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Model trained on a single static HDI dataset (historical, not auto-updated)	Predictions may not reflect the latest country statistics	High
2	Only three core indicators (life expectancy, schooling, GNI per capita) are used	Limits accuracy compared to the full multi-indicator HDI methodology	Medium
3	Flask app runs as a single local/hosted instance without load balancing	May slow down or fail under high concurrent user traffic	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask instance handling requests sequentially	Deploy using Gunicorn/uWSGI with multiple workers behind Nginx, and containerize with Docker for horizontal scaling on cloud platforms (AWS/Azure/Render)
2	Data Storage	Dataset loaded from a local CSV/Excel file at runtime	Migrate to a managed database (PostgreSQL/MySQL) or cloud data warehouse to support larger, continuously updated HDI datasets
3	Performance	Model loaded via pickle and inference done synchronously per request	Introduce model caching, batch inference, and asynchronous request handling; explore lighter models or ONNX conversion for faster inference
4	Security	Basic Flask app with no authentication layer	Add HTTPS/SSL, input validation and sanitization, rate limiting, and optional

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
			API-key based authentication for external integrations

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate live World Bank / UNDP APIs to auto-refresh country indicators	3 months post-launch	Keeps predictions current and reduces manual dataset updates
Phase 3	Add a comparative dashboard to visualize HDI trends across multiple countries and years	6 months post-launch	Enables richer analysis for researchers and policymakers
Phase 4	Expand model to include additional sub-indices (Gender Development Index, Inequality-adjusted HDI)	9-12 months post-launch	Provides a more holistic development assessment tool